Cilium Open-Source Dev Summit Part 1

Mar 23, 2026

Summary

Generative AI policy was discussed with a focus on human accountability and acceptable use, followed by the introduction of an MCS API proposal for cluster selection and an analysis of IP tunneling and mesh format limitations.

Generative AI Policy Draft
Discussion of the Generative AI policy draft focused on maintaining human accountability and ensuring contributors own their work, especially with reduced costs for submitting a pull request. Acceptable uses of AI include generating boilerplate code, but submitting content a contributor does not understand is unacceptable.

MCS API Cluster Selectors
A proposal for the Multi-Cluster Services API was presented, introducing Cluster Selectors and a State of Policy field to enhance service affinity across cluster groups. The key decision was to resolve conflicts in service export properties using an "oldest wins" strategy for global consistency.

Mesh Format and IP Tunneling
Issues with the existing mesh format include verbosity, size limits, and a lack of support for graceful termination, which results in instant traffic drops. A proposal for a version 2 format using compressed Endpoint Slices was suggested to align with load balancer ingestion and address the 1.5 megabyte etcd limit.

Next steps

- [ ] Someone in Hoge Veluwe (nl-ams-cla)'s Presentation will try to resolve the discussions on the generative AI policy document and roll it out more widely for wider feedback sometime early to mid this year.
- [ ] Someone in Hoge Veluwe (nl-ams-cla)'s Presentation will share the link to the generative AI policy document more widely if the audience is not already a committer.
- [ ] Someone in Hoge Veluwe (nl-ams-cla) will send a BPF test pull request that proves the current implementation does not work out of the box for the IP-in-IP scenario.
- [ ] Someone in Hoge Veluwe (nl-ams-cla) will add validation to set the limit in the next release and try to make the error message understandable.
- [ ] Someone in Hoge Veluwe (nl-ams-cla) will test the Go v2 implementation to see if it helps with performance.

Details

- **Generative AI Policy Introduction and Schedule Review**: The UCO EU psyllium developer summit 2026 commenced with a review of the rough schedule and logistical details, including the ability for remote presenters to use the GVC link (00:00:00). The initial discussion centered on the pressing nature of generative AI in the industry and the need to establish a project-wide policy on acceptable contributions. They acknowledged the ease of contribution facilitated by AI but must balance this with the project's high standards and the time constraints of maintainers (00:03:48) (00:11:18).
- **Discussion on Generative AI Policy Draft and Guiding Principles**: The current generative AI policy draft, initially drafted by Bill and later iterated upon, was presented as a non-official working document (00:03:48). A core guiding principle for the policy is maintaining human accountability, ensuring that humans remain central to the open-source endeavor and take responsibility for their changes (00:06:24). The policy aims to address the shifting balance where the cost of

submitting a pull request (PR) is now much cheaper, which is leading to an increased burden on reviewers (00:11:18).

- **AI and Contribution Quality**: The conversation shifted to focusing on contribution quality, suggesting that the project should prioritize high-quality contributions and shared maintenance burden over policing the specific tools used by contributors, such as AI (00:10:07) (00:14:32). The submission of a PR was historically a "proof of work," but that is no longer the case, which necessitates focusing on the quality and context provided within the contribution, such as detailed explanations in Git commit messages (00:10:07) (00:12:16). A proposal was mentioned that a focus on clear explanations and smaller, focused merge requests might be a preferable solution rather than strictly opposing AI-generated content (00:16:00).
- **PR Submission Expectations and Ownership**: Creating a PR, especially a large one, immediately creates an obligation and expectation of response time for reviewers (00:17:31). It was noted that while PRs are easier to create now, contributors are expected to own the code, maintain the continuous integration (CI), and look after its working status, which is often contrary to "drive-by" contributions enabled by AI. They determined that the next logical step was to review the concrete proposals for acceptable and unacceptable uses of generative AI (00:18:41).
- **Legal Guidance and AI Policy Scope**: The Linux Foundation provides guidance primarily focused on legal aspects, ensuring no contributions change the project's license or include copyrighted code incompatible with the project's licensing. The project's draft policy intends to go further by setting expectations for contributor behavior within the project without being overly prescriptive about specific tools (00:20:29). A key component of the policy is to allow maintainers to disengage from discussions or contributions if the interaction patterns are non-constructive or if the contributor is not actively involved (00:40:18).
- **Acceptable and Unacceptable AI Use Patterns**: Acceptable use includes using AI for learning the codebase, understanding complex parts, generating boilerplate code, reformatting existing content, or using it as inspiration for solutions (00:23:01). Unacceptable use includes responding to human-written responses with AI-generated text or submitting content the contributor does not understand (00:30:53) (00:33:10). The overall consensus is that a contributor must be willing to back up their work and be transparent if AI tools were used (00:25:43).

- **Security Risks and Generative AI in Workflows**: They noted that it is necessary to be very cautious about integrating generative AI into project workflows, citing recent exploits like the trivia and Klein project incidents, which used AI to distribute malicious binaries (00:41:22). They emphasized that individuals with repository write access must be mindful of how they grant access, particularly concerning tools that might use existing tokens or perform unintended actions (00:42:41).
- **Policy Rollout and Next Steps**: The generative AI policy draft was shared with the committer group for review and discussion, with the intent of resolving outstanding issues and rolling out the policy more widely for feedback in early to mid-year (00:46:26).
- **Presentation on MCS API and Cluster Selectors**: The meeting transitioned to the next topic, a presentation by someone in Hoge Veluwe (nl-ams-cla)'s Presentation on the MCS API and the proposal for new cluster selectors (00:47:40) (00:53:02). The existing service affinity only allows preference for local or any remote cluster, and the new proposal includes the addition of two fields: `cluster selectors` (a list to select groups of clusters) and `state of policy` (00:56:27). These selectors would use Cluster Property CRD, functioning as a key-value system attached to clusters for selection, similar to Kubernetes label selectors (00:58:14).
- **Cell Expression API Design and Cluster Selection**: someone in Hoge Veluwe (nl-ams-cla)'s Presentation discussed the design of a cell expression API, noting that on every selector there would only be one possible type. They explored what output would be desired from the cell expression, such as a label or a true/false predicate per cluster, to manage service properties across clusters (00:59:42). The current approach involves selecting clusters first based on an ordered preference, such as local cluster, then clusters in the same region, and then all clusters (01:01:14). This method dictates that cluster selection occurs before any service-related activities (01:03:00).
- **Diagramming and Design Consensus**: someone in Hoge Veluwe (nl-ams-cla)'s Presentation suggested that providing a diagram showing the relationship between regions and topology components could help form context for expected usage patterns (01:03:00). They confirmed that while the larger design direction has been mostly agreed upon following internal discussions, specific details still need to be finalized. The core process involves selecting all clusters and incorporating every endpoint size before service API actions occur (01:05:12).

- **Service Export, Import, and Global Consistency**: someone in Hoge Veluwe (nl-ams-cla)'s Presentation clarified the roles of service export and service import, noting that service export flags a service for availability in other clusters (01:05:12). Service imports are automatically generated by the Cluster Manager (CM), maintaining a global consistency where the same rules must be defined everywhere. Conflicts in properties are resolved using an "oldest wins" strategy (01:06:50).
- **External Controller and Placement Decisions**: someone in Hoge Veluwe (nl-ams-cla)'s Presentation introduced the concept of an external controller driving cluster decisions, suggesting that multi-cluster tooling that schedules resources could potentially drive service placement (01:09:08). They acknowledged that placing a placement decision object directly into a service might feel unusual, suggesting it might be better as a separate Custom Resource Definition (CRD) (01:10:32). The goal is to get close to traffic distribution, but without entering the service mesh domain (01:12:52).
- **Cluster Properties and Global Service Features**: someone in Hoge Veluwe (nl-ams-cla)'s Presentation explained that service export is the only writable field for a user, which is the only way to manage or stop the export process (01:12:52). They planned to transform the service import, which has a zero service attached, into a static list of cluster names that a load balancer would process. This approach would grant global service the added feature of being able to select clusters (01:15:18).
- **Consistency and Local Affinity in Cluster Selection**: Marco Iorio confirmed that cluster selectors on service exports must be consistent across all clusters, relying on the concept of favoring local clusters first. Someone in Hoge Veluwe (nl-ams-cla)'s Presentation confirmed this assumption, stating that any differences in selection rules for the same service would result in the "oldest wins" conflict resolution being applied consistently across all clusters (01:17:06). Marco Iorio inquired whether a simpler approach, like Kubernetes' service affinity, which prefers local, was considered (01:18:54).
- **Flexibility vs. Simplicity in Cluster Property Standardization**: someone in Hoge Veluwe (nl-ams-cla)'s Presentation stated that a simpler approach was discussed but noted limitations, such as the lack of standardization for cluster properties beyond two existing standards. Standardizing properties like region and continent, while potentially simplifying things, would limit the flexibility required by users who want to select based on custom properties, like country (01:18:54).

They expressed concern that standardizing every possible property, such as region and continent, would be challenging (01:20:13).

- **Cluster Properties for Network Policy and Operator Management**: someone in Hoge Veluwe (nl-ams-cla)'s Presentation suggested that cluster properties could also be used for network policy as a grouping mechanism. The operator would learn remote cluster properties and annotate the local service (01:21:58). Marco Iorio noted that changing a label or property would necessitate regenerating all identities, but acknowledged that these properties are typically static (01:24:25).
- **Handling IP in IP Tunneling with External Load Balancers**: someone in Hoge Veluwe (nl-ams-cla) presented a specific issue regarding IP in IP tunneling within an environment using an external Layer 4 Load Balancer (LB) running a Cilium agent. The current configuration uses BGP to announce the addresses (01:52:23). The problem arises when traffic targeting external traffic policy cluster nodes is forwarded to a node without the target pod, necessitating re-tunneling or SNAT (01:55:11).
- **Decapsulation Issues on Kubernetes Nodes**: someone in Hoge Veluwe (nl-ams-cla) detailed the flow of an IP in IP packet, noting that the first node must forward the packet, but the issue is that the destination IP in the forwarded packet ends up being the pod's IP, which is unexpected (01:56:27). On the second node, the traffic reaches the pod's eBPF program, which cannot correctly handle the IP in IP protocol (protocol number 4) and forwards it directly to the application, which is not the desired behavior (01:57:52).
- **Proposed Fix via IP Cache Lookup**: someone in Hoge Veluwe (nl-ams-cla) illustrated the current flow where the eBPF program passes the IP in IP traffic to the kernel for decapsulation because it doesn't know what to do with it (02:01:07). They proposed a simple change involving using a lookup from the IP cache to determine the correct backend, which should be the node IP instead of the pod IP, to correctly process the tunnel (02:02:48). This change would ensure the traffic is handled by the Cilium IP for device on the host node, allowing the pod to receive only the original inner packet (02:11:00).
- **Addressing Load Balancing and Tunnel Termination**: someone in Hoge Veluwe (nl-ams-cla) acknowledged that using external traffic policy local would avoid the extra hop but declined this approach due to the overhead of teaching the external LB about frequent pod movements (02:04:21). They confirmed that the tunnel needs to be terminated early to ensure network policies work and the client's source address is visible to the pod (02:05:38). Someone in Hoge Veluwe

(nl-ams-cla) concluded by stating they implemented a Proof of Concept (PC) handling the decapsulation on XDP and planned to send a test pull request to demonstrate the current functional limitation (02:15:23).

- **Continuation of the Previous Discussion and Scheduling**: someone in Hoge Veluwe (nl-ams-cla) indicated that they could potentially demonstrate a process live in the kernel if time permitted, acknowledging that another presentation was also scheduled. They mentioned that they had some topics prepared, but not a full presentation, and confirmed that lunch was scheduled for 12:00, advising participants to grab food quickly due to the small office and cafeteria (02:17:11).
- **Introduction to the Mesh Format and Existing Issues**: The discussion focused on a CFP (Call for Proposals) regarding the "mesh" format, which is used to share service data, backends, and global service information (02:19:52). Someone in Hoge Veluwe (nl-ams-cla) outlined issues with the current format, noting that it is verbose and its size is linearly dependent on the number of IPs and ports, compounded by dual-stack configurations, making it non-compact and reliant on JSON decoding (02:21:31).
- **Lack of Graceful Termination Support**: A critical issue is the absence of connection information in the current mesh format, meaning that when an endpoint is removed from the associated PF map (e.g., due to an endpoint becoming "not ready"), traffic drops instantly, which is problematic for graceful termination (02:21:31). Someone in Hoge Veluwe (nl-ams-cla) clarified that even if an endpoint is marked as terminating in the deployment specification, the existing format simply drops traffic (02:23:11).
- **Format Size Limitations and Distribution Concerns**: The verbose nature of the current format is problematic because etcd has a maximum value size limit of around 1.5 megabytes. This constraint limits services to a maximum of about 10,000 backends per service per cluster (02:24:40). Furthermore, distributing large format files to every agent in the mesh poses a concern for network load, with a projection showing that rolling out an update for 5,000 backends to 10,000 agents could easily exceed 100 gigabytes of traffic (02:27:30).
- **Impact of Churn and Rate Limiting on Updates**: someone in Hoge Veluwe (nl-ams-cla) noted that if an endpoint becomes "not ready," the existing cluster service must be updated, causing churn (02:28:48). Marco Iorio explained that aggressive rate limiting in the custom server coalesces updates, which significantly affects the number of updates generated during events like

introducing new conditions. They also noted that in cases of high churn, rate limiting might have a much lower effect (02:30:02).

- **Proposal for V2 Format using Endpoint Slices and Compression**: someone in Hoge Veluwe (nl-ams-cla) proposed a version 2 format that would embed the endpoint slice object to reduce data duplication and align with the existing load balancer ingestion path, potentially allowing for shared, faster code (02:37:31). To address verbosity, they proposed compressing the data, noting that initial experiments with one compression showed about 50 times compression (02:39:59).
- **Performance Disparity in Ingestion Paths**: Benchmark comparisons showed that on big amounts of endpoints, the cluster mesh ingestion path is significantly slower than the load balancer ingestion path (02:37:31) (02:52:42). Someone in Hoge Veluwe (nl-ams-cla) expressed concern that without aligning the formats and sharing code, any performance optimizations made to the load balancer would not benefit cluster mesh, which would remain a separate and less optimized path (02:57:29).
- **Trade-Offs of Restructuring versus Incremental Improvement**: Marco Iorio acknowledged the value of switching to a V2 endpoint-oriented approach but expressed hesitation due to the large shift required for a small team (02:46:25). They questioned whether an incremental path might be better or if completely reworking the structure to align with the endpoint slice structure would pay off in lower future maintenance costs (02:56:02) (02:59:06).
- **Need for User Data and Limit Communication**: someone in Hoge Veluwe (nl-ams-cla) and Marco Iorio discussed the difficulty in determining how much the current limitations affect users, suggesting that optimizing for services up to a certain size (e.g., 2,000 backends) might be a pragmatic path (02:51:09) (03:01:06). Someone in Hoge Veluwe (nl-ams-cla) emphasized the need to add internal validation checks and display clear error messages to users when they are about to exceed service limits, especially if those limits are decreased by adding new fields (03:04:19).
- **Conclusion of Discussion and Lunch Break**: The discussion concluded with an agreement on the importance of addressing the current limits and providing clear communication to users. The participants then broke for lunch, planning to resume in 45 minutes (03:07:40).

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00:00:00

Hoge Veluwe (nl-ams-cla)'s Presentation: and then all right okay um welcome everyone to the UCO EU psyllium developer summit 2026 thank you for coming and um this is the rough schedule and anyone who has the uh that GVC link can just use it to present so you don't have to come up here to the presenter thing to do the presentation and with that we'll get started with our first session So, um, so the schedule works very well, right? Yeah, we can swap things around if people are there's interest. I don't know if it's going to go that long, by the way. That that's just um the spare time at the end. So, was there a specific one of those topics or Maybe. Yeah. I think do you want to swap it if if we want to? There's consensus for data. So, um Okay. Which one's your slot? Okay.

00:01:19

Hoge Veluwe (nl-ams-cla)'s Presentation: I guess we'll just do this then. Did you want to be involved in that one? You need to read as well. Yeah. Oh, let's just keep it the same. We We'll just spend most time on the scale thing. Yeah. Okay. Marco. All right. So, maybe Joe, you're up first. Yep. Um
Marco Iorio: Hello, morning everyone.
Hoge Veluwe (nl-ams-cla)'s Presentation: just getting started. Um, Joe, you can just use my laptop if you want to like scroll through. I just uh I'm going to share this. Or do you want to share it on yours? It probably it's easier if you share yours. Yep. You should see this. Yeah. So, um, so obviously generative AI is a pressing topic in our industry.

00:03:48

Hoge Veluwe (nl-ams-cla)'s Presentation: A lot of people have a lot of quotes and

opinions about how should or should not be used. Um, so I think it's important for us to have a conversation about that and figure out how do we as a project um align on what we think is acceptable and um how do we balance the easy contribution which is becoming easier to some kind of contribution um but also need to balance that with sort of maintainer time that they want to put in. uh and the expectations for how contribution to the project should go. So, uh I guess you guys may have been involved in this uh discussions page on the community repository which has a whole bunch of different different discussions on different aspects of this whole um subject. If not, you're welcome to to browse and check out after this session as well. Um but uh through that discussion, Bill initially drafted um a generative AI policy. We had a bit of discussion there and then I've taken this into Google doc just to iterate a bit more and um this is sort of a current draft.

00:05:11

Hoge Veluwe (nl-ams-cla)'s Presentation: It's not the official policy. Still plenty of time to come and change things. um but does feel like we're getting more and more to a point where it's useful to have policy that says this is what we want to see about contributions and then we when we see contributions that don't meet those expectations we can say okay here's how we think that contributions should be should work within the project have make sure that we're we're kind of aligned um so yeah so I figured probably just makes sense to go through this document. Um it's not too long. Um overall the structure but acceptable use unacceptable use some discussion on transparency and attribution GCO and licensing legal aspects have covered u and we also have a guiding principle which I think we still want to iterate a bit more on maybe I'll start. So the the preface area so we're wanting to maintain the high quality and trustworthiness of contributions guidance. This is what we mean by when we say generate AI.

00:06:24

Hoge Veluwe (nl-ams-cla)'s Presentation: So it's all the large language model language model and the intent is for this policy to apply to everybody brand new contributors reviewers committers everybody. What I might do is actually just jump straight to guiding principles because I think that's maybe interesting us um I think there's a a core

idea here about human account accountability. Um so when people are making contributions you know open source is a very human endeavor humans trying to figure out what is the best way we want to solve certain problems. Um and uh we want to make sure that humans are still at the center of this. Um that humans are taking responsibility for those changes. What does that mean in terms of when you're communicating it's nice to have humanto human communication and not you know some you post a thoughtful response and somebody sends to a a text generator and sends back the response right? People don't like that. Um I'm I'm curious maybe this is jumping the gun like what is the project because we know this already happening right like people's discussion like the other end of the discussion is like not like what happens in that case like what what practically happens yeah what practically happens like you close the PR or something like what is the like we know because it's going to happen more often Yeah.

00:08:03

Hoge Veluwe (nl-ams-cla)'s Presentation: Yeah. Yeah. So I think this this um policy is intended to give us that and that answer there are some things that touch on exactly what would do with that. I think in practice what we have seen even without the policy is if somebody posts a significant amount of generated content on an issue or response on a pull request a lot of people will look at that and say well I recognize this generated by tool. I don't know if there's a human behind this discussion. I will disengage. I've seen that behavior. I've seen people explicitly request, hey, I've I've done it myself in certain environments. Um, but I think it's also there's a range of different contributions we see. So there's from one end which is the um kind of open claw type thing where PR shows up and even if you ask a question, there may or may not be a human driving that interaction.

00:08:59

Hoge Veluwe (nl-ams-cla)'s Presentation: um over to actually quite thoughtful discussions, but then maybe there people using AI tools to help with translation or to touch up their English, that kind of thing. Um and then what I what I found is that some contributors are maybe a little bit sheepish about that. It's like, oh, I just post something and then I expect it just gears into box contribution. Um but then they tend not to want

engage in that discussion and I find that's not a super healthy dynamic like right open source is about discussion and figuring out the best ways we do the best workshop one another figure out what works what doesn't you know poke holes and and collaborate right um but when I find even if somebody is using AI tools um but if they're engaging in the conversation and they're able to sort collaborate then then typically it's it ends up fine. I mean there's definitely some hurdles about like the expectations about how the the the tools are used but um I've seen that kind of go in a more productive direction.

00:10:07

Hoge Veluwe (nl-ams-cla)'s Presentation: So I think one of the goals here is not to say you know like AI good AI bad this like AI is a tool um people will use it um we don't police you know like oh why you using them or you know like or anything like that right um so yeah it's kind of more I mean in general I think one of the points Andre brought up relatively early was what we want to see is quality contribution and we want to share the burden of the making of the project. So uh you know we should be able to judge things based on on quality of whatever contribution. Yeah because I know like some for example curls like famously like close off like or something like hopefully. Yeah. So I think it's important to recognize is that there's a shift in the balance where in the past submitting a PR was proof of work often spent a significant amount of time thinking about the how to format and even just like mechanically putting the PR together to then send that contribution.

00:11:18

Hoge Veluwe (nl-ams-cla)'s Presentation: Um that's much much cheaper now. Um and people who have less context, less awareness about what the right way to do it even like in documentation or any any kind of area in the project. Um and so because that cost is much lower but review cost has not gone down that means that the shift in the balance of time. Yeah. Um so reviewers are feeling more over overloaded. So then we need to figure out okay one thing is can we make review process easier and that's one area we can explore. this policy doesn't really cover that too much but happy ideas on that. Um and the other question is well if contributors are freeing up time by spending less time on creating contribution in the first place are there other ways we can facilitate them to maintain or help help share the burden of the overall project? Okay, sorry I jumped I

jumped very briefly.

00:12:16

Hoge Veluwe (nl-ams-cla)'s Presentation: Well, yeah. Um so that's some of the high level points. Um there was a lot of projects. One of the things that caught my attention when you're talking about your tokens that translate to our projects that's not does not have time to give more data. So like if folks can come up with giving us more Be careful if that's one direction. What would you do with that? Basically the result of that is a comment that gives you more details on how could you say if a given issue gets assigned to a s now When you say context, you mean context. Oh, all right. All right. uh like context on maybe one of the practices we tended to have within the project particularly in the s repository itself is in your get commit messages explaining like oh I experienced this issue in this kind of production environment you know this combination of this factor and this factor and the fact that I scale and you know all these kind of why and like how context feeds into that's why I think this is the right solution to this problem and I think a lot of people like really like the way that you

00:14:32

Hoge Veluwe (nl-ams-cla)'s Presentation: can just do get log through some code find that original context that informs how to do better solution in the future and also helps to review like at least some text we're saying you're arguing here's a problem I need to solve and I've thought about it and I think this is a way to solve it bring that on if you talk about it like this it sounds to me a a lot more like uh you're not worried about whether it's AI generated but quality of it. All right. So uh actually when you said when someone said the note about Vim right you don't police if someone's using V don't police if someone's using uh LSP. So, but there's a uh format, there's lenses, there's some style things. I was wondering if maybe you were watching what's happening on another project. Maybe something those lines. So, the The reasoning behind that is give us enough data point. Yeah. Yeah.

00:16:00

Hoge Veluwe (nl-ams-cla)'s Presentation: So that brings me kind of to the point of view the solution might not be to kind of fight AI with AI. You know you would say I don't want issues that are very much unexplained or uh uncomprehensible huge comets then what do I want? I want smaller comets I don't I don't need it to not be AI. If I want smaller focused merge requests that are very much explained, that sounds to me maybe like uh might want is actually use the AI that is if you trust that tool in way. So, uh, it sounds to me like you would want a review agent maybe kind of to say, hey, this needs to be kind of maybe scoped to very particular part. There needs to be clear explanation. that this and um you know um kind of uh I feel like Joe is talking about some guidelines that are not really written or enforced on the call and this might be a way to do it if you trust the you know Yeah, I think like at least what I saw in some projects is like what I heard I'm hearing is like you can't just need a PR.

00:17:31

Hoge Veluwe (nl-ams-cla)'s Presentation: I think that that just I think that's what the point of that. Yeah. PR the PR is generated and then you basically just throw this like giant thing like at people and like that that probably I think the concept there is that the you you connect with the maintainers you agree as humans there is a problem here it should be solved and we have an idea about how to solve it. We agree on that. Discuss all that on the issue first. um because creating a particularly a large PR but creating any PR immediately creates a feeling of obligation or response time for reviewer and that's actually something we haven't formalized within resilient project but obviously anybody who sits on a PR for send a PR in two weeks there's no response like it feels feels bad so then as you know as you step up and you start to review in the project you kind of feel that well if I do better then I'll get a better response and I also So um I think with the issue the bar feels a little bit lower uh and the comprehension is probably a little bit easier than the large PR shows up.

00:18:41

Hoge Veluwe (nl-ams-cla)'s Presentation: Also, I guess you didn't touch the ownership of your plan to spit the code and get it merged, but there's an expectation I guess we

don't have this written down, but basically that you'll be an owner of code is part of you'll be looking after CI making sure that it's working. So personally with those drive by now it is easy to do. I'm like a signal that actually opposite. Yeah, your code and careerly they have to give us But you already have to trust them the first time. Yeah, the trust test for sure. So, um yeah, maybe the there's so many different aspects that we can go go into here, but uh maybe something that's useful is to go through like what we've put here is acceptable and unacceptable use. So we can at least like this is a concrete proposal that will have generate plenty of discussion by itself I think um and yeah maybe take a step back as well.

00:20:29

Hoge Veluwe (nl-ams-cla)'s Presentation: So Linux Foundation has provided some guidance overall that guidance is primarily focused around let's say legal issues and kind of paraphrase but it's you know you can't submit something that's going to change the license of the project submit under some license that is incompatible um and you can't knowly submit copyrighted code with that that with a license doesn't allow you to contri something that's not compatible 2 GGL project. Those are really the two core things that that come down from us from a learnings foundation NCF. Um, this is probably trying to go a little bit further to say here's what we're expecting people to do within the Zillian project. I'm trying not to take a prescriptive use this tool, don't use this tool. Yeah. Like some of those tell if they're using it, right? Like Oh, yeah. Yeah. Um Yeah. What you do in the privacy of your own ID is possible to tell.

00:21:43

Hoge Veluwe (nl-ams-cla)'s Presentation: Sure. Sure. Yeah. Exactly. Yeah. and and um to some degree like to some degree any any given policy is the degree to which you can enforce that people follow the policy is like you can't build it into everybody's ID you know whatever but um but like if you recognize a pattern you say I I don't think that the way that we're interacting in this is actually pushing us in the right direction um you know it's causing friction for whatever reason then we can look at the policy we can say okay is is this actually matching up with this uh this what we're kind of agreeing as desired contribution patterns and if not we can have a discussion about how do we improve that whether it's changing the contributor's behavior changing the policy you

know if necessary using that as a basis and so yeah again there's like such a range of contributions I think let's say that on the kind of floor end of crypto spectrum And touching on Martinez's point, if if you're submitting the PR in an entirely automated way, it's not obvious to me that you're actually adding any capacity project to then review maintain.

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Hoge Veluwe (nl-ams-cla)'s Presentation: We just have robot PR showing up and then it may or may not project. So like at that kind of level maybe want to say I mean it could be like there has to be a human somehow overseeing that and then helping us actually make sure this makes sense the project and typically like this is stuff that maintainers would do. So if containers are more likely to to do this stuff, they're overseeing it making sure the quality is right and trust the contribution that okay the spectrum maybe if it's a brand new contribution I mean for all we know it could be a supply chain seen that the last two weeks a I digress a little again. Um so um acceptable use and unacceptable use. The goals here are to say here are some patterns we've seen that work for people that typically don't create any sort of friction for others and then here's some other patterns unacceptable use where it's creating friction for the um and yeah we can I don't know how to pick on any given point but um so we've seen good patterns where people you know maybe you're unfamiliar with the codebase you want to learn get up to speed you need like a fancy grip to try and figure figure out how do you where is the policy code and how do I interact with this part of the the code base like learning about it getting getting to explain parts that you don't understand even if it's not 100% it might give you a kick start to figuring out how to contribute in that

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Hoge Veluwe (nl-ams-cla)'s Presentation: area a lot of the boiler plate kind of stuff like basic test framework uh table table driven tests or something that's exactly the same code every single go um some stuff like that is pretty pretty straightforward. Um yeah, reformatting existing content where you're not creating kind of brand new contribution sort of it's more on the toil end of the spectrum. Um I've seen also because it's these tools are trained on kind of a whole bunch of data from all over the place including yes um there may be like solutions that you want to develop and turns out somebody in

another project somewhere has developed or even discussed here are three different ways to solve this problem and so asking these tools to say hey here's a problem space what what exists out there and knowledge about this problem this that can be useful inspiration. So I think those things are probably I felt like those were probably relatively uncontroversial like and I'm not even sure as a project where it's safe to say don't do any of those things.

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Hoge Veluwe (nl-ams-cla)'s Presentation: It seems like it's probably beneficial if you you want to use it that way that's great. Um so this section also sort of I guess hints at how contributions good contributions look like you're involved in the so once you actually get to creating the PR itself you know you're involved in creating that entire contribution you're reviewing that content before you send it for others to look at right um you know you you're understand there's some discussion about the the take personal responsibility. I think that's probably a little bit too generic. Like there's meaning there that we're not capturing the documents. So I'll probably pause on that one for the moment. Um but the the concept is like if if you send us some contribution for uh for review and the reviewer says, "Oh, like why did you do it that way?" Or like, you know, what do you think is the right way to do that? It's not really okay to just say, "Oh, I don't know if the AI did it, you know, or like you know, this you are the one as the contributor creating that contribution. You're asking somebody to spend their time on it.

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Hoge Veluwe (nl-ams-cla)'s Presentation: Um, you should be willing to back it up. And like if you use AI tools, like that's fine. Like you should be upfront and just say, "Hey, I did this is why I think it's okay." And I think when you have those discussions, people will recognize that you are the one contributing, have thought about this problem, you think it's the right way to solve this problem or to write this content. Um I think mostly people are okay right from an perspective if you want to say that thing would make it to our PR is the goal to have something in our documentation to the documentation and maybe add a few blurs. You really understand? Oh my god. This was a big discussion actually yesterday with the Kubernetes the maintainers of um like there is one concrete proposal here which is the last one in the acceptable use section which is to report the use of AI

tools. The proposal here is this this um there's a somebody who's reasonably big in the AI space who's written this AI influence level years ago, which is like a scale of zero to five.

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Hoge Veluwe (nl-ams-cla)'s Presentation: How involved were you in the project from like you didn't use AI tools at all through to um you know, I kind of sneezed in the direction of an AI tool. Okay. Um my paraphrasing, but um so one of the proposals would be like in the PR template and say like, oh, please rate your you know, 0 to five. Here's here's the link. Um and but do you understand the the generative AI policy? I think is kind of on a on a similar tech to that. You think that might be useful like check like you read generative AI policy like you sign. Well, so one of the proposals actually yesterday was to have a CLA style process. Yeah. One of the things the tricky parts here is like I mean most of the contributions come from established community people in this room or like people who are like actively involved in the project right so I think for those people I mean I guess I guess if it's just like checking the box every time it's probably not that big of a deal but we don't want to optimize at least partly for that common use case.

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Hoge Veluwe (nl-ams-cla)'s Presentation: Um but then when it's people we don't recognize, we want to make sure that we're on the same page about what we're expecting contributor. Yeah. The the advantage of one of those like CLA like things is that you can't then in a discussion where maybe you bring up some of the unacceptable uses like well I didn't know. say, "Well, actually, you technically did go through that, right?" So, it's like you should know. So, that just kind of short circuits. You have any numbers how bad this I can give plenty of anecdotes. I don't have, you know, numbers. I mean, the other day I closed three hours, but the trend is The change definer. I don't receive that much. Um, so so maybe I'm a little little conscious of time. Yeah. little uh little late on this but maybe I think the unacceptable use is probably a little more controversial or like at least could warrant some discussion.

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Hoge Veluwe (nl-ams-cla)'s Presentation: So maybe if we can we could probably highlight we can go through those and if something yeah let's go triggers then like we can we can decide how much time spend on that. So one piece of feedback I've heard from quite quite a few uh committers people very involved in the review process and maintain is when somebody sends AI generated content in response as part of PR or issue description some sort of like direct communication they don't like that at a base level but I think it's um something is is lost in the whole contribution process when like for instance a maintainer might feel that uh I see your contribution. I read it. I I I think I understand what you're trying to do. I think it should be done in a different way. I spend 20 minutes writing, you know, I think this is how I think we should go. Um and when somebody takes that and gives it to AI and post the response back, the maintainer feels well, I could just do that myself.

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Hoge Veluwe (nl-ams-cla)'s Presentation: I don't know why I'm involved. Just adding sort of noise to the process. Yeah. And I think there's also some concern that people would generate a lot of text and they may not have read it themselves. There's a kind of this this feeling of like why should I read something that you haven't the time to write. So and like maybe even at a more concrete level I think in a lot of cases that text is it's sort of the best averaged Yes, the internet average internet average of what the response to this discussion should look like in the way that people interact and it's not necessarily highlighting what you're concerned about. And I think the the line between I think about this, I'm worried about this and I want to make sure we do this well versus like I haven't even thought about this at all, but here's a here's a text response. That that line is very unclear.

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Hoge Veluwe (nl-ams-cla)'s Presentation: It's not it's not it's not known like which problems of this this space you're aware of and trying to improve how much you're helping us to maintain that code. So the the first proposal here is is more or less like on a PR on an issue don't respond to a human response with an AI more or less although

these are not I'm reading it it doesn't look super controversial because even if you strip out the AI part of it it's still makes sense, right? Like don't submit something you don't understand. So maybe that's maybe that's a good like bar for the whole policy is to say if you take out the generative AI, it should actually just be what our current and document are. says we have not reviewed and tested explicitly say you understand isn't that the last line yes I okay yeah so these are not for Sure discreet I think still kind of forming exactly but I think the last point does cover that maybe that should be bold I mean one of the proposals was delete all the rest is the last point so one there's a spectrum of issues the easy ones are they comes from obviously content has science to say that this is easy.

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Hoge Veluwe (nl-ams-cla)'s Presentation: You're already all here already trained to detect that you can go and close that. But the hard end of the spectrum is someone took some time talking but I don't have enough gave some context of stuff the all looks on the face value of it looks face value of it looks correct but when you look at the comments they are not how a human like the use of I don't like you're using let's say you decided to different like if a human wrote that code you understand that okay this would be harder for review it so I'll try as much as I give my context on why I did that right and that doesn't exist and it's just generated like content and it's I there are situations where I I was super slow because it was not it was not like what I'm used these are the stuff where it would take like three four times longer for me to review because the descriptions are not have to read the code I might click right in there.

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Hoge Veluwe (nl-ams-cla)'s Presentation: But it's code like that. Although it's not after I understand all the code then I can translate what he said in this. I mean I I've seen seen furious contribution which alleged complex blocking issue was u which is maybe it's not the same case but it's it's a high bar like even if you recognize okay I think that this is probably AI tools there can still be a legitimate bug there um and so it is hard to to know how seriously you take that contribution. I know I think as a project we do the best work when you have kind of established trusted members who are ongoing have an ongoing contribution ongoing interest make sure the project goes well and then that provides

many many touch points for discussions to share understanding oh like when that person sends PRs they communicate in this way and that's fine um so I think overall I mean that's where I think the majority are. So maybe what you're concerned more with is a bit more kind of yeah someone's motivated to get started like That's very smart because they're actually giving a little bit more content of that.

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Hoge Veluwe (nl-ams-cla)'s Presentation: But I think like the typical push back, but again using the take LLM out of it is like if it's like confused like oh I fixed a bunch of locking prices like but I don't understand like that's a typical interaction. Can you please explain it? Maybe have a test that demonstrates like this actually a problem. So I think it probably not a new problem but maybe an easier. Yeah, I guess the test part is I've actually seen people that before you even read whatever can write a test that proves that maybe that's the test can be broken. I don't have to this is the worst part that so so yeah one one thought I have about this is like well with any infrastructure beh if if you have that like a very sort of positive feedback loop with the contributor you post a question they respond it's clearly like two humans like figuring out how this actually works. Usually, you can find a way there.

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Hoge Veluwe (nl-ams-cla)'s Presentation: Now, that can take quite a lot of time. So, one of the things I'm kind of looking to do with a policy like this is to say it's okay to sort of recognize this and try and figure out how to shortcut that process if you don't think the contributor contributing like actively themselves and involved and and working in a in a kind of positive positive way. like and it's always going to be a balance. But like I I think there are cases like there are much much simpler cases where you look at something you can tell it's AI. There may or may not be some value here but you feel obligated to spend a lot of time digging into it. You feel obligated to send a very nice response like ah sorry this and this and this and this. Um and that is so timeconuming it's not necessarily going to actually get to solution we want. In a lot of cases, I've seen as soon as you send like a one question back to a AI generated PR, the author disappears and there's no response at all.

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Hoge Veluwe (nl-ams-cla)'s Presentation: So I I'd rather have this situation where we can at least engage a little bit and see is there a person actually actively driving this contribution? Yes. Okay. Then we can continue this discussion and actually figure out how to work together to solve it. If not, let's shortcut this and say, "Okay, there's way to not spend your time on it." Basically, time check, Joe. Maybe um anything else I think you want to highlight in this. If anybody has a legal department who uh feels confident about posting a comment about then uh it would be interesting to to have that discussion. Obviously none of us are assume none of us are so a glance of that. Um the only other thing I would say um more focused on um people who have privileges within the project is kind of this section here. So this is not part of the current proposed policy, but there's a section that here's a bunch of other ideas in light of some of the recent exploits we've seen like the trivia trivia repo just disappeared overnight.

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Hoge Veluwe (nl-ams-cla)'s Presentation: Uh Klein project uh suddenly started distributed binary automatically. Um there are ways that we could technically integrate generative AI into our workflows um that have the potential to comp. So I think as for anybody who has a right access to any sort of repository I think we need to really be thinking about how we're grunting access to don't have a good understanding what's bounding any sort of right access I think here I'm nominally putting out like just don't provide access at all that if you have a GitHub token don't grant flash access tool run in a DM whatever I think this is a problem generative AI has to has to solve over there but as a project we should probably also very very careful about how so like the coin project one for instance was is a AI triage bot issues that was used to poison cash was used to publish binaries And so by filing an issue that prompt injection in there, you could actually release a binary that has super scary like it's the the bar is actually quite low.

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Hoge Veluwe (nl-ams-cla)'s Presentation: like this is not um oh I have to but you know

as reviewers for instance right so you pull down an ER if there's an agent MD file in that and then you run that through tool you're just giving access yeah super obvious this thing. You're already using that. Now the possibility of these tools using your exist is it's a real possible risk. So like I I don't even think CLI you can like an employee token kind of I think technically you could probably create a personal access token that only privileges probably the the page I'm thinking of is a GitHub page you go to the personal access tokens create a new token and say here's all the permissions you can have and there's a bunch of check boxes I think it has read by default and you check what other privileges you can give it maybe I haven't checked for sure I mean this is also partly why I race it's like this is not something that's built into standard would be an interesting Um and yeah, if there's actually anything we feel like GitHub can do better, like almost proxy summit discussion there, I know GitHub runs a contributor summit.

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Hoge Veluwe (nl-ams-cla)'s Presentation: They have a maintainer community area where a lot of people give feedback. They are receiving feedback. They're trying to adapt. They recognize a bunch of these problems and they are concerned about it. Do they know how to solve it? I don't know. But uh they're certainly aware we're in trying. But uh yeah, I think we're probably running a little over time. So um wrap it up there. I'm around. All right. So currently what's happening is I sent this to the committer group because the committer group is ultimately the one this not there's actually a bunch of discussion like I hit it to avoid getting too much into the you know the minutiae of various different discussions but um I can share this link more widely if you're not already a committer um I just figure go through the committ group first at least try to get a first pass on do we think these points are good set of points on where the policy should be at any major kind of started with more or less Bill and I drafting it and sent it to a few extra people with different perspectives on

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Hoge Veluwe (nl-ams-cla)'s Presentation: AI to get some initial review. I've now sent it to committer group for wider review. Um the discussion is still ongoing. the next couple of weeks I'll try to resolve those things and then roll it out a bit more widely for more a wider feedback hopefully sometime early mid this year or so we can have a policy where

cool I think every project is navigating we'll see what happens okay next topic um We have presenting. Yeah. IP DSR. Um, let me Oh, are you already logged in? Oh, I have to catch seems to be. No, I cannot like Oh, weird. Can you just just bounce out of that and then try it again? No, no, no, no, no. You don't. Oh, weird, dude. Wait, which I wonder. Well, you can you can share a tab.

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Hoge Veluwe (nl-ams-cla)'s Presentation: No, share a tab. Oh my. Yeah, open it in tab and just share that. You have Yeah, but try you can also try plugging in here if you want. There's uh there's this this stuff here. There's a HDMI if you can you can just plug in. Yeah, physically. here. Yeah, thank you. Hopefully this works better. Okay. I mean, it's a new thing, isn't it? They're trying to figure out how to get it. Yeah. But it's kind of funny because there's like, you know, these like mirrors. It's like, but what were you actually doing like with all this token? Like what happened there about like you create? Oh yeah. Is it working? I'm not sure what uh is there I mean how do I check if it's should be visible here or I mean uh adjust menu present.

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Hoge Veluwe (nl-ams-cla)'s Presentation: Oh, did No, that did work. Nothing should go. Okay. So I like two kind of um so the first one is MCS API. So it's actually kept in mostly But I mean doing this mostly for CM. So but there are also also other implementations that I might not be one. So yeah it's kind of very small presentation of what the API looks like and also a brief mention of um what then the implementation would like. Uh so basically right now we have an an addition called service affinity uh which helps you to say I want to prefer only the local cluster or any remote cluster. Um so the fairing local poster is really useful. The remote one I'm not sure if makes that mention setups we don't have many agency well choose besides every other remote cluster um and there was already some kind of issue in the past. It was closed by the ste mostly.

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Hoge Veluwe (nl-ams-cla)'s Presentation: Um yes, someone was interested into um

making it more complete. Um I think he was proposing here to the name. He was proposing to also have multiple ones. Um and uh yeah, I think M was involved was really that the contri didn't finish up the PR and um I'm trying to a bit address all this topic. It's only right now otherwise it's fine up with the support. Um and in short it could like that. So well this is the current proposal. It might change a bit over time. Uh but basically you have two uh fields that are added. Uh the main one is cluster selectors and then you get a list and every um items select some sort of group of clusters over on your cluster mesh or setup in not about C. Um and basically we take uh the first one that has any radian points of rad um in light of like um the um prevail standard service basically distribution field.

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Hoge Veluwe (nl-ams-cla)'s Presentation: Yeah. Yes. Um and uh there is also um state of policy which just make this list open ended or closed ended. Um so that like if you don't match anything after it just gone right there. Um but this is um this was added quite recently and it will not stick. But yeah, the main idea um and the main thing that we take as a source is um something called cluster property which is a CRD by signal cluster and it's essentially like a glorified key value that is attached to cluster. So kind of a level system using that you could select our cluster like network policy kind of and we have a true type of well selection I don't know if critter is the best word but um yeah we have level selector which is literally the the struct in kubernetes with match levels match expression like that uh which would the other cluster property and then we have some other things but those are more like long-term plan.

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Hoge Veluwe (nl-ams-cla)'s Presentation: Um so first like the cell expression thing uh but um there are multiple like API design target that we could do and uh um I never design the second do they already have cell on that or is that okay so it's like something new yeah yeah it's like um basically on every selector you would have only one type possible right and uh wow we only target the first on right now looks really interesting but I don't know exactly how to design that API and yeah like I'm just wondering what you would get what you would want popping out of the cell expression like a label that then you like how would it essentially you have the all the properties of all the cluster of

exporting services and then have a way to well figure out the output of the clusters I see and uh would be like a like a true false predicate per cl yeah this is one way the other way is to just pass all cluster together but there are some cell cost aspect that I don't really know how to measure well so what's the use case is it like there's this property and this property or this other property and some monging of those properties I guess using the cell function some label like so you could imagine just hitting a different label.

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Hoge Veluwe (nl-ams-cla)'s Presentation: Yeah. Yeah. It's like does it does it degenerate to labels in which case labels? Um so wait maybe I can show so this is a few example with like level set might be more understandable that so um what I not sure I mentioned but uh there is some concept of adding some this special value that get replaced and um yeah other than that. It's the same thing as your usual um level selector strict and all of that and same as in the importing cluster. Um like yeah logically is that um and basically this expression means I want uh only my local cluster then I want all the cluster in my region and then I want all the cluster in my um and so if you have any rad yes It's an order. Yeah. What happens if you Yeah, very good question. Um, so yeah, didn't expect that. Um so the the goal of that is that we choose cluster and and in the choosing cluster then whatever service things happens.

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Hoge Veluwe (nl-ams-cla)'s Presentation: Um so you can choose cluster and then you can have traffic same AC and have um like prefer same a between those cluster and you could have actually multi cluster within same yeah if you prefer local selection. Yeah, that's yeah. So, we always select cluster first and do not care about anything below. But this was um a conversation that we had in uh for simplicity we chose that way first but later we could have some kind of take traffic into a control or something along those lines. Yeah. But this is getting more complex. So just to like have it maybe configurable later if someone's going to view the use case probably something like some some diagram somewhere just showing the relationship between the a what you're expecting these region topologies component like specifically for this stuff like yeah because if we say like we're expecting people to use like this overall topology this overall topology I mean that can help to kind of form out of the context there's no diagram but it's like well

I mean I don't know just just a suggestion but at least I'm not what's the reaction to the by the way like what's the discuss So, so like it was kind of I didn't send the cat first.

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Hoge Veluwe (nl-ams-cla)'s Presentation: I like have like I don't know four or five discussion and sort of shape the how the design looks like. So there there are details in that case that are not viewed yet, right? But the larger like direction is mostly agreed. Okay, got it. Um but yeah basically the point is that we select the all clusters and we take every endpoint size in that and then some service API things happen no this multic so for me it's like when I do the service I guess that's the point when I want the whole thing of the service export. I see like bunch of service providers and then there are consumers of those services they do the set the preferences on oh so I guess that's what you do with this same as important right uh maybe I can some because like the so export is just to say I want to flag a service to be exported in other clusters. Uh but um the service import are like automatically generated by CM.

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Hoge Veluwe (nl-ams-cla)'s Presentation: So what I'm trying to say like when I do the okay it's automatically generated but it was not like manually I would say inside this service import that okay this no you you cannot like um like API has a concept of global consistency like when you export something the same across So that's also why the same as in kind of exist. So you basically have to define the same rule everywhere. Mhm. And then there is a a conflict strategy and basically the oldest win. Um and uh yeah this then get resolved when you are importing the and for this for the for the right yeah for the import so like the um like you export the same properties and uh then if there are any you put in the oldest wins and it always leads up to the on every service. I think the import is generated right like it's it's not user so it has to be consistent ideally also happens automatically yeah imports are generated like they're derived objects yeah sorry I was kind of focused on the API that um yeah so Basically what happens is that you get those service export you find it everywhere you need to export something and then um everywhere on every service import it should yeah and um also just to mention that we have

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Hoge Veluwe (nl-ams-cla)'s Presentation: like um third type which is oh this is fa first kind of the cell. But um one of the cool use case might be to have some kind of external controller driving the decisions of the clusters. Um and there are um for instance in multicluster tooling there are something that schedule resource or pods on various clusters and those tool could also says I want to drive the service um like do do you the placement decision CRD is that something that's already defined or yeah uh yeah this is for like those of I see I see but there is which tool specifically or is there a standard in I think there is like karma and stuff like that okay but I I'm not a huge expert in that area okay but the the I think it was the karma maintainer that brought up this and it quite cool um and um but we don't know how it would shape up right So like placement decision is only a scheduling thing. So it might feel weird to put it directly into service.

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Hoge Veluwe (nl-ams-cla)'s Presentation: So it might be another C on the night. But basically the the idea that we might want to explore is like an external controller just providing implementation the the service. There's a lot of overlap how you want. Let's say if you were to combine traffic service well like the the things that are serviceive in the scope but it's like um explicitly processed after and all the service mesh except not really cuz we don't want to go in that service mesh area like we don't want to take those and kind but like like the also on gateway API I don't think they want to address those kind of things they they have no way to say in a service I want to those cluster and stuff like that. Um and I think there was like multiple discussion around that on GitHub and this like not really want this everyone well my inter my inter that but if they wanted to be that I don't know how that covers But this is a first class to say okay this is down for some reason.

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Hoge Veluwe (nl-ams-cla)'s Presentation: Yeah. But basically our goal is to get very close to traffic distribution which effect a bit that point um to service meshes and stuff

like that. So we aim to do the same but I don't know how it So is this saying like um I will only export the service the first one if it no like you um so you export all of them with like preference. So the service export is always attached to a service and it will always export but this is just for the because service export is the only writable field by a user. So it's the only way we can stop that um because you cannot write to service import as a user. It's completely generated and we don't want to add more fields to service because it would be weird that get some sex to like any like that is not used by default something only concerns for yeah I think I didn't fully catch that either yeah yeah that that can you maybe share how this relates to classic service because seems to like be somehow uh similar to global service you merge all back ends right yeah uh your cluster mesh all the if if I have a service and the same name spaces cluster that service is annotated by global service merges all back ends right so how is this so my my And is this so

01:15:18

Hoge Veluwe (nl-ams-cla)'s Presentation: basically so before I only shown you like service export and so service export at the end of the day service import on clusters that will this then there is always a zero service attached to it like various annotation that we can add like clusters and my plan is to transform this into something like this, like just a static list of cluster name and then load balancer would go over those and the first one that has rad and um so there's two aspect there. So well first we need to have cluster properties in code so that we can um actually have the crystal working and if we do that we can only incorporate that into MCPI controller that live on the and then have a very simple list that are processed by agents. the select back ends mechanism for the new um new dish state TV integration and all of that. And um yeah the the other thing is that with that like global service gain a nice feature along the way which is that they can like cluster and if they want to have a more high level syndrome related they can go there.

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Hoge Veluwe (nl-ams-cla)'s Presentation: So fine that's valid for some comments.
Marco Iorio: Uh, one quick question.
Hoge Veluwe (nl-ams-cla)'s Presentation: Yeah.

Marco Iorio: Sorry if you already discussed that. I'm only hearing like 50 or 60% so I might have missed that. But um, like so my understanding here is that the cluster selectors given that they are on the service export they need to match on all these service exports, right? Like they they need to be consistent across all clusters. So um by design you cannot specify different selectors on different clusters but you rely on the fact that you probably want to favor um like local local first basically. So first maybe by region then by area or whatever and so you use this same as importer or whatever some some kind of wild card to express this that concept. Is that uh is like my assumption correct
Hoge Veluwe (nl-ams-cla)'s Presentation: Yeah, exactly correct.
Marco Iorio: here.
Hoge Veluwe (nl-ams-cla)'s Presentation: Like if you have any difference and like for the same everywhere.
Marco Iorio: Maybe one followup question. So um I I I I haven't follow super closely but there was this trend in Kubernetes especially for services to try to simplify as much as possible the concept of um like service affinity.

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Marco Iorio: So they moved from a more generic uh label based approach to
Hoge Veluwe (nl-ams-cla)'s Presentation: Okay.
Marco Iorio: like only prefer local basically and then fall back to this or that. Um so was this discussed as well? Do I keep it very simple? Uh prefer uh local local region things like that compared to uh having that um all this flexibility of you know you can specify by label selector or even cell or custom logic which can be arbitrarily complex.
Hoge Veluwe (nl-ams-cla)'s Presentation: Yeah, we we did that but well we discussed there are multiple issues like the first one is that there is no uh there's only two standard cluster properties. Uh so we will have to standardize like region, continent and stuff like that. And uh also people that have um that want that would want to select our custom properties like a country for like
Marco Iorio: Yeah. Yeah. I totally agree with that. It was more like not not saying that would be better.

01:20:13

Hoge Veluwe (nl-ams-cla)'s Presentation: Yeah.
Marco Iorio: It's more like whether that was considered like with the idea of keeping things super simple, super straightforward, which on the other hand limits the flexibility.
Hoge Veluwe (nl-ams-cla)'s Presentation: Well, yeah. I think we also we're afraid that okay, we need like region, continent, we need I don't know something in between and and many other things. So it's it's like hard to figure out what everything should come up and grow kind of not but kind of in the large side. I mean I I can probably think of five on the spot. So is it actually similar at the end of the day?
Marco Iorio: Yeah.
Hoge Veluwe (nl-ams-cla)'s Presentation: Yeah.
Marco Iorio: Yeah.
Hoge Veluwe (nl-ams-cla)'s Presentation: Like like different properties or locality. Yeah. Like what I'm interpreting from the slide like there's two use cases. one is about um yeah I mean I also add some other example for um let's say you have um I don't know you have some scaring on your cluster like this is a good machine and you want to have first I don't like so yeah And also with cell there are kind of nice things that might be able to do like I have 10 in I want to buy this thing or this is

01:21:58

Hoge Veluwe (nl-ams-cla)'s Presentation: like I I don't know what people will use for but kind you're you're kept includes to sell stuff or is that like a future. Yeah, it's okay. Somewhere along those we want to explore that area, but I see. Okay. I don't think it probably somewhere. Okay. Um just a quick time check. Yeah. So well I'm done. So cluster property would like be used to select clusters but will also be network policy because it's a kind of grouping things I was quickly mentioning last year at the last I started that that yeah you could have something like that on so another kind levels and that they will be processed more locally. So watch that every which is like level which can add miniatures if they are added. So possibly knowing like before this what I was thinking is um individual nodes don't actually need to know what cluster property because the operator is learning that for all the remote clusters and then annotating the service with the service name then the services you're thinking I don't have to prop two nodes to like but I think this is saying actually each silly node has to have some notion of this because policies later only on the local it will be

01:24:25

Hoge Veluwe (nl-ams-cla)'s Presentation: naturally shared like you add it to the local yeah it's like an But but I guess the reason why within the same cluster that allocating those identities to pull that so I guess with the operator manage identities that actually yes but but the good thing is it's not like a we force every node everywhere in the world. Yeah. Well, Only the owner
Marco Iorio: Oh, sorry. I just wanted to mention this comes with the usual cut of changing the label or changing the property would mean regenerating all identities regardless of how that's done. So that's obviously a problematic operation. But I think on the other hand, it like they feel pretty static to me. So it's something that gets configured once and then used more or less forever, but it needs to be taken into account.
Hoge Veluwe (nl-ams-cla)'s Presentation: Yeah. Yeah. I'm totally sick of like making like the
Marco Iorio: So
Hoge Veluwe (nl-ams-cla)'s Presentation: agent would probably not just discover that the CR exist and automatically enable it, but we probably have to opt in first.

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Hoge Veluwe (nl-ams-cla)'s Presentation: And there will probably some warnings there. Maybe some default um inclusion in what the levels are. It helps remember how it's detecting tricky part is that if somebody does want to enable that then by itself I guess that's always a balance and Oh, right. But did that ever go anywhere? No, it does work. No, it just like turned up. I mean, there's no guarantee. If if the turn ends up hitting, you know, Oh, yeah. It's those when it's it's not processing one by one. It's like processing all at the same time. There's no like Yeah, there's no structured. Okay, interesting. I'd say if anyone wants to do that today and cluster like that and if it's like all the But like it's this is when you like have it's when you select labels that are not yet added or just by default on the So if it's part of the HD labels the label label add new label removed a label from a cluster name any that falls within that section.

01:28:46

Hoge Veluwe (nl-ams-cla)'s Presentation: And if you have any kind of policy maybe not if you update the policies to match both the old and changes I mean that that can help but then on an individual say pod basis the first agent that then allocates that pod Um, first it needs to allocate that identity. Then for everything to work, it needs to propagate that identity to every other node that might policy with that. And then those nodes actually need to calculate that policy calculation policy and only then you can guarantee that you could actually swap that on that node identity new identity and not a fix. Otherwise, like if you just immediately switch the identity, there's no guarantee anyone else understands what that identity is. they may start dropping eventually things consistent and catch up. So process that but then yeah unless you hit the limit but what what um Andre said like if you if you change some let's say name level that you never say on description if you just if you don't select.

01:30:16

Hoge Veluwe (nl-ams-cla)'s Presentation: Okay, it's part of the security. Okay. So like if you solve this well it actuallyes or we can just say yeah it's really it's error right now. I mean I think that's the feedback we've g pretty pretty clearly like I don't realize like like you change an space label I think there was one case where it changed let's say for like the entire cluster or something like a label or something it's not used in network policy but you suddenly added label to every single pod everywhere and then because it was considered relevant from whatever the defaults were set up like that means that every single Basically I guess that the initial design was that should be stat I think it was the idea was you have security relevant labels if the user understands what that is they know oh I'm going to enforce policies based on thing then they can customize that and that work in practice it's fairly I I think the problem is kubernetes never gave any kind of structure to label Yeah, we don't know which ones actually are.

01:32:03

Hoge Veluwe (nl-ams-cla)'s Presentation: Well, I mean, I haven't stepped too many toes in that area yet, so I don't know if I'm super confident about that. like just I mean just copying the name um being able to watch the study feel I don't know how to do what

you just described but to be honest like this operation is already pos you're still it's on you that that operation can but if you're building abstractions that allow people to like deploy a CRD and then make this all happen that's even easier today it's just a CM deploy with okay even even today the thing about changing having some automation changes namespaces labels from the entire cluster like that's the same problem. somebody thinks oh I'm just innocently adding thisformational tag like you know maintainer equals joint you know suddenly I reop to reassure you a bit like I was always pushing for property to be mostly static but I I was imagining that if you change some property that you doesn't really matter everywhere you won't that much.

01:33:45

Hoge Veluwe (nl-ams-cla)'s Presentation: I think it's good just to like just like state that where I'm from. This was designed with this in mind. You try to do dynamic be somebody I think that's a fair place to start otherwise you have to tackle very Okay. Um I think we'll come back at 11 for the schedule and then um folks on the call I'm going to hop off the call so we can just truncate the recording so and then but we will be back at 11.

Session ended after 01:34:24

01:52:23

Hoge Veluwe (nl-ams-cla): All right. Um, you ready to present? Uh, I can't try using Okay, another shot. Oh, do you want to You're starting on silicon, right? Yeah. Yeah. Are you presenting presenting here? I I'll just test it. Oh, yeah. You're right. Okay. Thanks. Okay. Uh, never mind. Um, okay. very specific topic uh IP IP.

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Hoge Veluwe (nl-ams-cla): So IP in IP um uh in our case from the high level not doing anything but uh in our environment we have uh external LB which uh uh which is actually running a selium agent in the class you know the the old uh standalone L4 LB mode

which uh in current Selium code, you can still uh run that. You just like would enable Kubernetes false and that's still possible. So we've been running that since 1.13 uh pretty successfully and uh then just load balance the traffic between our nodes for example extra proxies uh or those can be Kubernetes nodes. So, uh, this uses, uh, a BGP to announce the uh, do we have Where's the microphone? Any Can they hear us? Yes. Uh, folks on the call can hear this, right? Yeah, there's like a ton of microphones. Can't see them. Oh, okay. Uh, so yeah. Uh, the the thing is that we have the IP tunnels here.

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Hoge Veluwe (nl-ams-cla): uh and then in case the the when you're using uh external traffic policy cluster right you are targeting all the nodes in the cluster not only the nodes that host uh one of the pots behind that service. So uh there can be a case where the traffic ends up on this node which then has to either uh source net the traffic or using uh like BSR do another form of tunneling to get the traffic to another node. Right? So this is the pretty standard e easy um uh setup and uh we if we talk about IPs and how the how the uh source addresses or or the IP addresses look like. So uh the first IP IP tunnel you have the outer IP is the external LB. The inner uh or the destination outer IP is the first node. uh the inner is the the original client request. So it's client address the service address. So this note uh then needs to forward forward the packet again. So imagine we we have only the IP IP.

01:56:27

Hoge Veluwe (nl-ams-cla): So the the the outer source address uh is modified. So instead of the external LB you have the the the nodes address and uh this probably should be the uh the second node address so that the traffic ends up ends up here and is decapsulated. So here in the inner IP header you still have the original packet. Uh now if if we have imagine we have this uh configuration of slim agent running on the Kubernetes uh on those nodes. So uh the native XDP hook uh for the acceleration this patch I want the IP you've seen that we have the endpoint routes and we want the IP to be terminated right the pots cannot be capsulate the IPI the pod is for example the classic engines or or API or whatever user application uh so we want the selium to decapsulate the traffic not forwarded to the to the pod itself. So here's here this is the like dump of the T uh TCP dump communication. So this is on the on the first first node.

01:57:52

Hoge Veluwe (nl-ams-cla): So there's no part running on that note and you can see that uh the packet arrives on uh let's simplify it e uh it gets sent to the forwarded to the IP device to decapsulate traffic. So here we are uh handling the the inner packet. Uh it gets sent out by the same device out to to reach the uh the second node with with the pot here. Take a closer look. Uh there is the port IP here which might be uh unexpected. Uh so what happens and it's it always been there. uh is when the traffic uh ends up on the on the second node with the pot uh there's a you know the destination is the pot IP so it gets forwarded to the pot which you know what's happening is like the first node is doing the load balancing decision right yes Kubernetes yeah so when it uh reaches the part B ppf uh program uh this is the first thing that can happen uh because the the code you know it cannot uh really handle the IP correctly but there's oh I'm not doing anything go back sorry uh yeah so I noticed that there is this pretty new flag which uh uh which allows to uh let the the unknown tunnel protocols uh go through the code and uh that helps but then uh obviously it

01:59:44

Hoge Veluwe (nl-ams-cla): reaches the the application or the you know the the the name space isolated port uh which still doesn't know what the protocol number four is which is the hip and this is what uh gets received by the first node that that only forwarded or lob balance to the the original packet. Uh which is not really how we want this to work and uh but it's how it is right now. Um yeah, I think that feature is not in Yeah. Yeah. Uh so I I understand that and uh I would like if we can maybe talk about uh if this is ready to be uh like used in in Kubernetes maybe all of you can can share what what you're using uh if it's SNAT for the for the um LB mode or if it's BXL or or Geneive um because seems to me this is pretty like um easy to use and it could work you know. So if we take a close close look at what happens this is the first node that only forwards forwards the traffic.

02:01:07

Hoge Veluwe (nl-ams-cla): Uh this is not external OB. No no no no no it's it's the it's the first uh it's the Kubernetes node that that one uh here the bot one. So uh let's so if I zoom in uh this is basically what happens uh there's the first the BPF on the E0 which tries to look into the uh L4 ports so the TCP ports but as this is handling the IP IP so the outer IP protocol is IP IP uh currently it doesn't know what to do with it so it moves it passes that to the kernel. So there's a return context act. Okay. Kernel uh sees that this is IP and passes that to the IP device. Yeah, pretty uh okay. And uh during the all the function calls, you know, there's this uh tail call into the the part that is supposed to send out the packet. There is the uh backend address passed via the context buffer maybe uh yeah it gets used for the for uh filling the the bites for the for the uh outer destination IP address and uh yeah simply similar thing happens happens here but as the destination is the pot IP uh there is no forward ired to the uh IP device.

02:02:48

Hoge Veluwe (nl-ams-cla): So yeah, we've seen what happens. So effectively what's happening is is it's just it doesn't know it's IP. It's just it's a packet arrives at a node port and then translate it and send it on and it happens. Basically there's some like discrepancy in the understanding between the external load balance from personal that package and I I'll show you it's uh there is very simple uh change how uh how both words can work uh how both scenarios and it's in this part you know if uh if we for example use this uh look up from the IP cache uh which uh should have information about the the back end whether uh what's the node that hosts you know the back end and if that was successful and we we use that then the the traffic would look differently and it would work you know because instead of the pot IP I would have the node IP. Mhm. Uh you know the same thing would happen initially on the on the XP hook but then as the destination would be the note IP uh we would have the same IP IP in there which would correctly process that and uh successfully you know forwarded only the the inner packet.

02:04:21

Hoge Veluwe (nl-ams-cla): I I I guess if we go back to your first like yeah for the uh

images my my like why don't you want to teach the external OB about the exact placement of so you mean use like external traffic policy local uh yeah technically that's what class balances because then you would avoid this extra hop I understand Um and then I'm not saying it's uh we cannot do that. We we could but uh teaching that would uh I mean you know if the pot moved between the nodes like frequently uh you would have like many events that the external LB would need to process and there could be times where it sends the packet to the node but the pot is already on a different one. Um, I mean, okay. So, that's one. Yeah. I don't want to change your I mean, but I know that I already like uh I've been thinking about the the the same thing. Yeah. Sorry. Yeah.

02:05:38

Hoge Veluwe (nl-ams-cla): We're getting to the if you zoom zoom out. Yeah. Why don't you terminate the tunnel like earlier? So, you get the packet because Yeah. The thing is I want uh the network policies to work. So I want the uh the source uh address of the client. I want want it uh on the pod, right? So that I can but but I'm saying that the from the external holding to the node one this is basically what Okay, I see. So you want to still keep So the question I have here is is like it seems like the the packet is targeted at a node port there but the packet is actually not it's it's like an IP IP. Yeah, but it's it's not actually intended as like the inner packet is intended to go to the service but then the outer packet is just delivered to a node that then knows how to do the thing. And the problem here is as I understand it is because that IP packet is directed towards the service.

02:06:40

Hoge Veluwe (nl-ams-cla): Zillian thinks. Oh, I need to do service translation for this outer packet rather than knowing, oh no, this is just an IP packet IP IP that I need to decap and then actually do. Yeah, because I think if you do the termination at the node one before passing the tunnel passing through the program. Yeah. Yeah. You will have what? Uh so you need to keep the source and destination right in the program. Mhm. So if you attach one program before take out the packet change the size like as a general statement when we're doing IP decor header uh stays the same. I was thinking like outer like IP. So you don't care basically about the outer like outer header at the point one just a transport delivery. Yeah. Yeah. So you could basically drop it before processing. I

could do that but still uh the it would end up the the same because uh imagine this node one would try to uh select uh back end where where it wants to send the traffic to and uh imagine it will be this this one and they would put the uh the code would put the pot ids into pot IP into into here which is fine no because it's that's how the non ports and

02:08:29

Hoge Veluwe (nl-ams-cla): IP. Yeah, it works for for this uh for this part because the the nodes are you know we we we have the uh the PPF program for the for the node port. So so the selium from net death you know is attached to the E0 uh as well as the selium i84 device. uh they both have the same same program attached. But uh if the destination is not the node itself uh the the kernel will simply like directed directly to the pot and not to the to the device which uh which is the paral let me show you uh I mean it's definitely weird because like at that point you're assuming the network knows how to route the traffic for all the pods as well. Yes. If if the destination like typically in a tunnel kind of different but the concern I would have actually is the inner IP destination at the final destination should be a concrete pod IP. It shouldn't be a service because if it's a service you could end up running yet another note.

02:09:44

Hoge Veluwe (nl-ams-cla): I mean maybe that's okay like you just round robin it until it finds a you know a node that actually deliver but it's not the yeah it's getting more and more inefficient in the worst possible case now maybe that's not so there is something else then that's okay I haven't touched that code in a while but node port with IP and does it work yeah that works first fine No, the external the external LB works beautifully fine. it it correctly creates the IP APK IP and as I said we've using we've used that uh for a couple of years now and work just fine but now uh we are trying because we did some changes in the infrastructure to have the BSR working uh and you know all the IP spoofing mechanisms that are there we we made it to work and uh just lately we We we tried to uh reconfigure our solution to to use the IP uh in the kernes as well and yeah it did not work uh out of box. So started digging.

02:11:00

Hoge Veluwe (nl-ams-cla): So basically uh yeah let me just just finish up this part because uh yeah if we did this changes then you know uh if the destination uh in the alter IP header is the node IP then you know uh all all the things work and we end up having uh yeah this was before so this only for the uh illustration and this is after. So there is uh the same upper part to the the first node one uh it's you know the same only the destination IP is uh switch for the node node uh two and on the on that node that hosts the pot that there is this selium IP for um in that picture that decapsulates the the IP packet and the the pot correctly uh you know receives the the original request. Um so yeah, so uh these are the changes I I did. So basically uh it does look up from the IP cache and uh for the external uh LB usually the the IP cache has nothing uh relevant for the for the back end.

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Hoge Veluwe (nl-ams-cla): So from external LLB point of view the the back end is one one of the nodes and there's no information about the tunnel end point etc because there's no Kubernetes that no no controller that that fills that information and in Kubernetes context as so this is the pot uh and that has those information. So maybe we we could have like uh this we could enforce the the lookup lookup uh in both cases and in the Kubernetes case you would uh use the you know as as the back end you would use the the the tunnel endpoint so so node so the node IP and in the external LB you know you wouldn't look up anything so you would just stick with with the with the backend address maybe. So yeah uh so uh maybe to to uh get the whole picture. So the external LB yeah this is the important part here enable K8s false and this is how we uh currently configure the the LB. So there is the endpoints array with the with the address of the node.

02:13:38

Hoge Veluwe (nl-ams-cla): I mean there could be like node name, maybe host name, maybe uh zone information, but uh none of these will uh will uh you know fill up the the IP cache with with these informations. So in my opinion, in my opinion that uh it could work. Yeah. Yeah. My my one comment is that that load balancing code somewhat

overload it. In the past we did this custom tailoring. I think if you add this if you change the backup format you might break others. Yeah. Other scenarios. And is it 19? Yeah. I'm still quite surprised why it doesn't work. uh you terminate the tunnel in the very beginning at the node one and you act as if the request came directly from the client to this because the pity is that there's no test for that for for this scenario there's uh few maybe months ago uh I talked to Julian he said that uh uh some of his his colleagues uh just recently ly added uh IP test but that tests uh it tests the the external LB part so the encapsulation and not the receiving part you know not the decapsulation uh as you can see here uh this is maybe once again uh focusing on how how the IP IP gets handled so if for example I would

02:15:23

Hoge Veluwe (nl-ams-cla): want to do everything on XDP uh with the current state but uh it doesn't do anything, you know, it just looks into the packet. The the the head points uh points at the outer uh IP and it doesn't do anything. But maybe there we could. And the question is whether we want whether we want to do we want to handle that or you really just want to leave that uh a little bit slower but you know leave it on the kernel to do the deapsulation. Um you know all this could all be handled on in the XDP in my opinion. I I want to connect with the other talk about data path very nicely combines you use. Yay. No, because what I'm still trying to push from my idea that you could do early step before execute and that's the perfect case for um true true. Well, yeah. So uh to sum up uh um yeah uh I did some some PC that uh that handle handles it all in uh on XDP uh which works uh but yeah uh let's see let's stay in touch later maybe the first thing I can do is to send a test p request BPF test that proves that it does not work out of the box.

02:17:11

Hoge Veluwe (nl-ams-cla): Maybe uh we can continue the discussion on that. Yeah. Yeah. I guess like uh just vanilla port. Yeah. What what happens with the maybe I can if we have time I'll I can show you live in our can you know maybe maybe if there's time okay uh yeah that's cool I think you also have another presentation no not really we have some uh thoughts on uh yeah like having Oh no yeah no no it's yeah okay we have some some some some topics but we uh not prepare that all Yeah. It's in this room, right? Yes.

Yes. They So, the lunch will be at 12. Yeah, this office is pretty small, so I think the cafeteria is like not very big. So, Maybe grab wrap it up. That's the one that you join. I saw you join me. No, but it doesn't look like Oh, actually we can uh differenties which terminate the use programs there.

02:19:52

Hoge Veluwe (nl-ams-cla): Uh actually we we kernel to do encapsulation. We already used that uh thought you had and I was like terminating the tunnel before selium touches the touches the packet. We already do that on on setup but um without any PF uh yeah we let caramel to do that and then use the PC with the yeah program in this case before very early. Yeah. Yeah. Cool. Okay. Yes. Yes. Yeah. No worries. Um so this is about um a CFP I created on that iterated between um so it's called mesh and it returns the data format that we use to share service in mesh all the back ends all the behind what's global service is essentially and uh well they are some kind of issues with that format. So this is what kind of looks like today. So um well at the bottom you see some that is no longer used because you have some you have some um features that was not there.

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Hoge Veluwe (nl-ams-cla): I think it's related to the fact that you could not see on the VM and have a feature or something that was a few weeks ago, but it's not super important. Um, the major thing is that here on the back end, you see that we have all the part info which address not very visible because I only had one. And also if you start to have multiple ports it would be here. You could have like two and then have another IP that uh maps. And so essentially the size of that script is driven by um is linear to the number of IP and the number of ports. Also if you have draw stack it's multiply by two on your number of back ends and all of that. Plus it's JSON decoding so well but it's not a very compact binary format and all that so it's very um the second point is that there is no connection info whatsoever. Um so it means that if I understood correctly the context about um all the load balancer stuff uh when you remove that from here it will essentially remove it from the PF map and I think it would like somewhat instantly depending also on the configuration like drop your traffic coming going there.

02:23:11

Hoge Veluwe (nl-ams-cla): Um so yeah if your end points become not ready um the traffic just kind of drop um which isn't great if you do some want to have some graceful termination and have a lot of your deployments and uh yeah you expect things to work like does basically um So can I have one question? Yes. Uh does it not work in a way that where you have a uh graceful terminating terminating period uh on the you know the deployment specification. Uh is it maybe there is uh when the when the pot switches from false to uh from ready up to to not ready does that not mark the pot as uh as terminating and the traffic stops going to that pod only the new traffic doesn't it work like that I thought that well you don't have any condition info so it just drops the same Oh, in case you mean in case the part is like remote. Yes. Yeah. If you if you have some um service affinity local you see this effect very much less I guess.

02:24:40

Hoge Veluwe (nl-ams-cla): So basically the the cluster cluster one point of view is that all the remote parts are ready no matter what. Yes. As long as the uh connection between the LCDs is wait. Yeah. and we filter um like parts that do not match this like export time. So when the operator exported to this but yeah this is like one of the big point with this making it not great. Yeah. And so the things that well not about like so condition so coming back on the fact that is being verbose um as you probably know there is like some max limits on the value size in it's around 1.5 megabytes and um so today if you have a service with around two because remember it's um linearly the size is linear with also the number of ports. Um I estimate that you could have a max of 10,000 backends. Um so it's per service per cluster not all the cluster from on from on the service.

02:25:59

Hoge Veluwe (nl-ams-cla): Um and potentially if you want to add more fields to that cluster service and keep it like the same way uh we could like reduce that number further. uh for instance there is like two things that I might want to add which are

conditions for the like terminating logic which I just explained and potentially endpoint name which I could explain later but it like would further reduce some code around things. Um there is one thing that makes it okay to have this kind of limitation is that um as far as I understood all the um back ends of all the service in the map which limited to 64,000 by default uh which you could increase I think for instance in GK it's increased like that so you could have more um and the other thing is that um in Kubernetes um Like the upstream is rated for 5,000 nodes. So if you have one B per nodes, it should fit that except if you have less stack, you can go more than that. If well around that is the max limit um because yeah 5,000 by two around 10,000 limits.

02:27:30

Hoge Veluwe (nl-ams-cla): Um so this is kind of okay. Uh but we are kind of concerned and we have to take care when potentially evolving that format overall. Um but the adjacent adjacency to that um is that we kind of also want to keep that um script kind of low in term of size uh because you have to distribute it to every agent in all your mesh. Uh so here I did some kind of napkin math. If you have like 5,000 back ends, it's around. So that size and let's say you want to correate to 10,000 agents. So let's say you have 11 with 1,000 node each. Um you would have 10,000 remotes endpoint slice the DDS. They're just like calculating number of like mailing all this. Yeah. So yeah, the nin map is around 7.5 GB and if you roll out the same like if you have a deployment driving this and you roll it out, I don't exactly know but it would probably be more than 10 or so updates.

02:28:48

Hoge Veluwe (nl-ams-cla): Yeah. So easily more than 100 GM just to roll out. it's for every node. So, um I don't know how big of a deal is it, but still um and yeah, we have we are essentially in the same situation as when currently switch to slice essentially, but we are still on the thing. I mean, yeah, you mentioned about like graceful termination kind of stuff. Yeah. How are you thinking about those updates in terms of how many updates are you expecting? What kind of churn or like I feel like there's probably some some element of how how dynamic you are with pods spinning up and down affects the ability for us to distribute those changes. So as as of right now like if you start excluding the end points here um it would um essentally do like um so if a pods become not ready for some reason you would have to update the the cluster service the existing one. So it

already does the churn.

02:30:02

Hoge Veluwe (nl-ams-cla): If we add directly without doing anything the condition info, it will probably add more because you have I guess more boolean than just on and off something like that. I don't really know to which point but yeah it's like something like that. Um was does this answer your question? Okay.
Marco Iorio: Uh I guess maybe one thing to mention one reason why it's sort of difficult to foresee
Hoge Veluwe (nl-ams-cla): Yes.
Marco Iorio: the exact number of updates or um that would be generated with any kind of event is that rate limiting plays a big role here. Um by default we have fairly aggressive rate limiting um in the custom SAP server uh which is also meant to sort of coalesence bunch of updates together. Um so that would um probably affect here like that would change also the number of updates in case of um there are conditions and things like that. So even introducing conditions that would change definitely the number of updates in case uh of low churn in case of high churn that would probably have much lower effects because of rate limiting then rate limiting is obviously configurable.

02:31:35

Marco Iorio: So it depends on how you configure it. Um you would have different um different results basically at the end of the day.
Hoge Veluwe (nl-ams-cla): I didn't quite catch everything because there was like some noise and all that, but yeah, the Yeah, are sure and I we don't see no point maybe there could be we had because we've seen that in uh in the past that we we had some that were switching PS like very frequently and if hund of these replicas doing the game like that's like disaster, you know, propagate each second, you know, it could flip from true to false and Yeah. Well, you can already be scary. Well, you can already have that essentially like on the services, you might probably have issues. And if you start looking closely at the um things like your feel like concretely what would be kind of interesting to know there is like one one is like getting a a good sense of here's here's turn scale and turn targets and then maybe having like overwhelming agents or something like that and then metrics that signal that users can know oh I can set and I can plan and say I

expect this much and then if it goes over your trip say oh like this is the condition you're in and now

02:33:45

Hoge Veluwe (nl-ams-cla): it probably is not as evolved as you are explaining right now but there is like on the is like a separate It's for everything. Yeah, it's right. Yeah, if you Well, that could help. It's it's an idea of where we could like I I may be thinking a little but I mean so it's uh it's a bas on every so you have I don't on the service and it's like this amount and you have many comp maybe this could be like off to the user Because in context if you have like multi- tenant cluster mesh that we have and there was there would be some like not uh you know some user that wouldn't know what he does exactly you know he could blow up the whole thing but in case this was a single tenant it would be his you know his problem I don't just mean but Yeah. And eventually Yeah. Yeah. Like I guess the the implications with UPS affecting how long it takes and if you constantly overload that any other operation but if you go over it will like call us updates on export time.

02:35:25

Hoge Veluwe (nl-ams-cla): So like if you have 10 updates in the queue on the same just one but it doesn't do that import show that in the very next slide actually um so the other issue about is the the time it takes to inest and here the code path is actually very different from like just me perspective and of the balancer. Um and well this matrix was updated after improved by on already. Uh but yeah here um so if we compare those two things because like because the so this is based on the benchmark of the load balancer. So it's like not end to end or something but just the insertion by many times and I think it takes the um well have fake client on I'm not I don't remember but it's not just something along the lines of we have state DB you have structure just like a whole bunch of them like state by itself as fast as units then yeah kind of looks like that's doesn't represent a large amount like the bottleneck well I basically I reproduce that benchmark but with data and those are u like the same thing uh and well I I had two mod that's benchmark so first uh like doing the whole insertion like I consider that I don't connect but I

02:37:31

Hoge Veluwe (nl-ams-cla): I like and I do like and one tricky to have directly object um and so to be fair with the benchmark here which I think doesn't include the putting on server I just compared those two and you see here like on big amount of end points it's like way slower and uh on the top lifeline like for instance if you have 5,000 back you areing around 36 millcond without including any network that and there is also no racing when you so it means that if you have the 5,000 back ends and you are updating more frequently than 30 millisecond or so you could stack up and have like updates way in the past and I reproduced that before optimized it with 5,000 back end up every 100 millcond I had to cla and I printed those level in the two cluster and one was like getting well the export was on time I could have like so but it's in right um so yeah that's all the problems from things um I do have like some kind of proposal um open ids basically I don't know if you have question first on existing issues let's go um so what's I would essentially propose to the um v2 of that format so well introduce a new format instead of like trying to update one um is to just embed the size object um

02:39:59

Hoge Veluwe (nl-ams-cla): this has um so two main advantages the first one is that you have less data duplication because the PS I don't know which one you have in mind the subject but they have the definition for every so you don't have this implication back end. So it's a slightly better format in that sense and also well the load balancer uh code that ingest from Kubernetes already use. So I think we could like very easily try to have a shared injesting pipeline and have like later maintenance and make both elbow at the same time and again 500 times and there's also because it's uh quite faster might not be actually needed. There's many performance benefit to the existing balance. Um and to fix the B aspect, um I was going to just compressing um I did some um experiment with one compress and it was around 50 times compressions with the like thousands or so back ends. Um I didn't do it with this proposal. So I I don't know exactly what one is on but it should be quite high number and um part of that um I was also doing some benchmark and it's around five times lower CPU wise um than just like decoding so not accounting the state in session and all that but yeah it's about 5% more than

02:42:18

Hoge Veluwe (nl-ams-cla): Um so this is not ideal performance wise because you don't have a chunk and with a chunk well you don't have to and decompress everything at once. Uh but there are two or three reason that we might not do that. Uh first we can have uh this decoding very easily paralized by it's already there. There is go v2 that might not help. I'm not sure because I've seen like some benchmarks saying faster and some reporting on go that it's not. So hopefully it is but I might test it out. Um and also because well we don't have any checking today so um it's the same portion probably change right now. Um and um yeah that's that's that was the proposal. Um there is one kind of caveat to that uh which if we add the end subject and we don't we include condition there will be more as we discussed bit earlier we also um so if we want to go forward Without um getting yet into the details um we might consider to just like feature flag those so that we have exactly the number the same number of updates as before and then try to uh really study like I what happens in updates if I have this flag on or off something like that.

02:44:16

Hoge Veluwe (nl-ams-cla): Um so yeah that could be a perform I think and yeah overall I'm kind of open other ideas. Uh the one the one thing that um I think I'm kind of attached to is that we have some similar script as endpoint size. So like like um we can just evolve with the source format and some process format that might evolve afterwards which was the case today essentially. So like um the the trick that we share today is I think something like eight years ago or so when was first introduced that made sense in the load balancer context and now it has evolved several time when a few years ago um but yeah Aside than that, um I could see I don't know another in format I explored bus but it was a bit complex because if you want to inspect then the values have to share the schema and I made some tooling around it and make the end to end test on strip more complex implement and all that. But yeah, part of that is fancy encoding that would really help a lot and very it okay I mean I would love to ready, you know, to be part of the uh of those structures because that might help a lot.

02:46:25

Hoge Veluwe (nl-ams-cla): Yeah. Um I think the flagging from a like if we're aligning flagging allow existing we're not participating in this transition.
Marco Iorio: Yeah. Um, so I guess few few thoughts. So, thanks a lot for the presentation. Like very good points in my opinion. Um, I I think I I kind of wonder if the solution to some could be like on the one end, okay, let let me phrase it differently. Um, on the one hand, I like this idea of going for a V2 solution. on the other I'm struggling a bit at you know it's fairly large shift uh for a very small team currently so that's you know kind of kind of we'll will take time and so on I kind of wonder whether there can be a more incremental path where we can you know gain um gain improvements along the line without doing a full shift uh but on the other And it's more like where do we want where what do we see as the end goal? Um and I guess to me there the main question is I can totally see the goal the the value of switching to a more endpoint oriented approach uh you know to better align with upstream.

02:48:11

Marco Iorio: I'm bit on the fence on whether it's better to have this single service with the list of all the endpoint slices or whether to have like you know kind of endpoint slice like uh entries within K the KB store each has advantages and disadvantages kind of currently we like putting all together means definitely the service is unbounded which is not rate like it's the only resource currently we have there which has loose size is not bounded. So that's um that's kind of one of the problems the one you mentioned here we could hit limits and also in terms of the amount of data that needs to be propagated. So that would steer towards splitting and having separate employee slices. On the other end, now we have this good property of we do patching based on the service itself and one service update consumes a single uh slot in terms of QPS while having endpoint slices if you had a lot of them then it would change the distribution in terms of how updates are performed. Um to be fair like any update in a service most likely corresponds to

02:49:29

Hoge Veluwe (nl-ams-cla): Yes.

Marco Iorio: an one or two updates in the endpoints themselves like in terms of pots um which also consume uh QPS slots because of like one pod will probably start another will be removed. So from that point of view, it's also something we could consider in terms of how we distribute uh the the update slots, how we do coallesencing and so on because we can be super fast at updating the service, but then we might uh struggle with keeping up with the updates in terms of the endpoints within the K store uh structure. So that corresponds to Celium endpoints or Celium member slices. Um because like if you have 1k pots which are being churned that corresponds to 2k at least 2k uh operations there and even if we super optimize the service path that would not mean that you are able to reach the service faster or you are able to really target those back ends faster because you are still potentially behind with those other uh 2k updates. So to me there are multiple things um that kind of play a role here and they are kind of tricky to uh like waiting the vantages on the one end because that can affect negatively potentially the other ingestion paths and so on.

02:51:09

Marco Iorio: Um so I don't really have a conclusion here. Uh one aspect which is bit fuzzy to me is what users expect in terms of scale. Um because like to my understanding the largest services in Kubernetes tend to be ones for um post network pots maybe demo sets and so on. If you have a 5k nodes cluster then you have a five 5k back ends. I'm not really sure um in a cluster mesh scenario what is like a real world limit on how many back ends you want to use to to have for a service.
Hoge Veluwe (nl-ams-cla): That's
Marco Iorio: Does it really make sense to have a super large service which is shared across cluster or is the common approach more to have like a
Hoge Veluwe (nl-ams-cla): quit.
Marco Iorio: few which may mean hundreds of back ends maybe across multiple clusters but that that to me is an open point. Not sure if someone in this room uh has has opinions, has feedback, but I think that would be super valuable in my opinion to drive like some of these uh decisions based on whether we want to really focus on high scale services, whether we it's much better to optimize, you know, uh lower scale services and so on.

02:52:42

Marco Iorio: And probably like the the final point is on the performance. Uh I my my feeling is there's probably some lowing fluids. There's definitely too much difference between the Kubernetes ingestion path and the cluster mesh inestion path. don't think that can be explained only by you know having um not having endpoint slices especially because uh up to 100 there should be no difference any like even in kubernetes that corresponds to a single endpoint slice I think if you go back to the numbers even that case was like yeah uh it's still 70 times slower so this uh to me that's one of the main signals that the load balancer logic was really optimized while cluster mesh was sort of okay we need to make it work with this the switch but um and it works but it definitely isn't performing enough um so I like I wouldn't say that simply switching to v2 two will magically fix everything here because to me this 70% uh 70 times slower thing for 100 back ends is probably not fixed by the switch uh because that maps to a single endpoint slice.

02:54:16

Marco Iorio: So it's still probably one update um and one up like one update here and one update there. It's probably just that the paths are totally different and they are totally differently
Hoge Veluwe (nl-ams-cla): Yeah. Yeah.
Marco Iorio: optimized.
Hoge Veluwe (nl-ams-cla): But like if we if we were to get the input test format like not saying that uh having like separate object just having process some place you just plug it in the the pipeline have some shell code. So gaining the performance that they just made like vacation stuff.
Marco Iorio: Not sure I got your your point. Sorry. uh issues with with the code but I I I like my understanding is you are saying um we would be able to share more of the code path and that would automatically help. I agree to a certain degree. Um there there will be for sure certain parts which need to be different because cluster mesh has some extra logic for you know preferring um I I don't remember exactly with the new implementation maybe the filtering is done afterwards but there is still some some logic which needs to be done and some some special parts which are specific to go mesh because of how how things work.

02:56:02

Marco Iorio: Um, so I'm not totally sure dear uh which um which amount of gain we could get. Um I think we could get uh significant amount even with the current structure. Um but again it's matter of like where to put the effort whether to improve try to improve what we have or just try to switch to something else
Hoge Veluwe (nl-ams-cla): just one like this like churn.
Marco Iorio: and
Hoge Veluwe (nl-ams-cla): So, it's I didn't actually say it, but um in the Oh, in the crystal mesh code, what I'm doing is I think I I'm just reinserting the same thing. Wait, no, I think I'm changing one back end. And in the load balancer, I think I'm reinserting the same str. Oh, wait. No, in the customiz code, I'm inserting the same thing. And in the load balancer test, I'm changing only one end points in each service. Um, so basically the load balancer, if I remember correctly, does slightly more stuff. Um but this is like to say I have 1,00 service I'm gening only one endpoint what happens uh and not I'm creating service from scratch because if you are creating the service from scratch um I think it's much more consistent

02:57:29

Marco Iorio: Yeah,
Hoge Veluwe (nl-ams-cla): between the two
Marco Iorio: like my my main point is it's it feels kind of wrong that it takes that much time to basically do nothing because you're saying you're not even changing one back end. So it's it's probably we are doing a lot of work that should not be done um just because the the the path there is not optimized.
Hoge Veluwe (nl-ams-cla): Yeah,
Marco Iorio: Amen.
Hoge Veluwe (nl-ams-cla): but I I think one of the point is like maintenance like If we keep those two path aside, um if keeps continuing support optimizing the balancer refactor, which is great by the way,
Marco Iorio: Yes.
Hoge Veluwe (nl-ams-cla): cluster mesh would still be probably a sign kind of and not benefit those performance benefits.
Marco Iorio: Yes, I I agree with you. Um that that's why I was saying like there's a bit of trade-off to me whether it like we want to have to optimize kind of what's existing today which sort of requires less work uh in terms of changing the the overall structure but it

has this like obviously means keeping to separate ingestion paths and so on with so that that the improvements in one not necessarily benefit fit the other or or maybe we can still make them um symmetric.

02:59:06

Marco Iorio: Maybe we do some prepro uh some pre-processing and then we feed it into the upstream load balancer logic. I don't know. Um just run some random thoughts or whether we want to go full in with um let's rework completely the structure. let's align cluster mesh to what uh like to the endpoint slice um structure and then benefit from there. So we pay maybe more at the beginning to do the switch and potentially with the goal of paying less in the future in terms of maintenance and so on. So to to me that's one of the main tradeoffs here.
Hoge Veluwe (nl-ams-cla): Okay. But because I I think If we well if you if we start to like add those if we we start to add like those fields that even if we like filter flag condition stuff like that we would like keep increasing the same strength um and we can like I don't know maybe if we say we don't expect service to be more than I don't maybe thousand parts or something but can be a fine thing I guess we just won't have like much your service overall but maybe with the end point and all of that ago that's not really relevant anyway cannot inest those fast enough and it's It's very important

03:01:06

Marco Iorio: Sorry, not sure if that was a question. I totally missed it. Um,
Hoge Veluwe (nl-ams-cla): like I I was essentially saying um we could say uh we optimize for I don't know up to 2,000 back ends and then um after that it's not really supported because the selium endpoint thing that you mentioned uh which we would not be able to process fast enough anyway. Um yeah that that could be a buff forward because we don't I mean I I don't have the matrix around but we're like the size I also made that version of the but it should be I don't know around maybe 200 3,000 kilob might be okay. This was not the
Marco Iorio: Yeah. No,
Hoge Veluwe (nl-ams-cla): question.
Marco Iorio: but again, it was not like I didn't understand your question was like I I I I'm hearing only some pieces of what you are saying.

Hoge Veluwe (nl-ams-cla): Okay.
Marco Iorio: So,
Hoge Veluwe (nl-ams-cla): Sorry.
Marco Iorio: it's more like a communication in terms of like, you know, uh call problem.

03:02:25

Marco Iorio: But, um like again,
Hoge Veluwe (nl-ams-cla): What is
Marco Iorio: don't don't get me wrong. I don't I'm not saying we should not go for your proposal.
Hoge Veluwe (nl-ams-cla): it?
Marco Iorio: I think it has lot of benefits. Um I'm I'm kind of struggling at probably one of the main points that we discussed in the CFP. I'm really struggling at seeing um how much this is a painoint for users currently and how much this will become a painoint um for users in the sort of near future which means it's difficult to me to see how like how much effort we should put into um like planning for these optimizing uh for maybe large large services uh compared to optimizing or improving correctness uh for for smaller
Hoge Veluwe (nl-ams-cla): So I mean just like my main agenda is to
Marco Iorio: services.
Hoge Veluwe (nl-ams-cla): add like I started to say oh we don't have oh if I just to add voice like a lot. I see. So um and I started from here and hey we're kind of limited if we go that a bit right you would reduce the but maybe this is fine metrics for this but we don't have metric we don't have but um Yeah, maybe we could stop there.

03:04:19

Hoge Veluwe (nl-ams-cla): We could even like Yeah, I mean the condition would probably your flag the employ name. I don't really know that number because for instance this I added quite recently like the zone info but this same zone works and then I realized a few months that oh this has decreased the max number right everything you have is going to decrease one concept would be like validation just I guess this is a I don't know like if you pick a number and then suddenly people stop complaining you know to some degree it's a little bit of a um yeah it seems like something what I'm hearing is like we need at least from my experience on the production side like we do

see not for cluster mesh but people do put it like some rare people do put that it's it's not like it doesn't exist. It's just probably very uncommon. I guess the flip side is also you put a limit and people will figure out differently. Yeah. So maybe you chase them off of that, but at least it would be good to have some end user data.

03:05:46

Hoge Veluwe (nl-ams-cla): I think on it sounds like in this area some end user data just on like how many period that sounds like I don't really know how to do that but maybe we could add in the values like max crystal mesh and the com like we experienced this with just like let's say like in kubernetes like At the very least, you should have some error message very easy to understand because like I think when Kubernetes in the past, it was just like things would just stop. Yeah, I'm just like like, "Oh, I didn't expect that to happen." No one knows what I never actually witnessed, but I guess it could be a weird error message. Yeah, like if it's like understand it, but right now it like I don't know what happened. But the thing is if people are relying on I don't know something right below this limit like I don't know 8,000 9,000 something add some stuff they would just like that's fine I mean it's not fun but like what what I mean to say is like yeah people who are involved in that process will help to shape what what the code should actually do right and And I think if you're a user in a particularly like high scale or particularly detailed thing that requires weird and custom thing um I mean like let's say we roll this out in 120 or something 120 comes out there still 119 still 18 so yes they try to upgrade hopefully they do it sooner than later they find out

03:07:40

Hoge Veluwe (nl-ams-cla): this is the limit and say okay like how can we work together I wouldn't trust that that they do it sooner or later we usually do it later than everybody else But uh but like to to some degree there's only so much at the core of stream. Yeah. It's like the problem of like oh I hit this problem we have a common user set and then you have like this long tail like okay the weirder the things are that you do the more I would like to see like it's not that we don't want to solve your problems but like we need community involvement. So like should we say I like let's say we I don't know something in between 1,00 4,000 and make like things out in like a really way and yeah then maybe a flag to like say that you can also like in the upgrade like reflight checks you could say

like first it checks through your resources say like hey this is actually going to start to trip this limit and then like even before they start migrating like as part of that initial transition like oh I actually cannot then they don't end up in this you know disastrous situation but they at least know I okay so I think this would be good to get us like like just at this limit in the next release and try to make the message as like the

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